Lab 7: Fuzzing for Input Validation Bugs

L6_L7_46 - Photo Print Order (photo_print_order.py, test_photo_print_order.py)

Step 2: Run the original script

```
$ python photo_print_order.py
extra_print_charge('4x6', 30, 'matte', False) -> 40
extra_print_charge('4x6', 80, 'canvas', True) -> 770
extra_print_charge('4x6', nan, 'matte', False) -> nan
extra_print_charge('4x6', 25.5, 'matte', False) -> 22.0
extra_print_charge('4x6', 25, 'sepia', False) -> 35
extra_print_charge(['4x6'], 25, 'matte', False) -> TypeError: cannot use 'list' as a dict key (unhashable type: 'list')
extra_print_charge('4x6', None, 'matte', False) -> TypeError: '<' not supported between instances of 'NoneType' and
    'int'
order_valid(10, 100, False) -> True
order_valid(10, -5, False) -> True
order_valid(-1, 10, False) -> True
order_valid(nan, 10, False) -> True
order_valid(None, 10, False) -> TypeError: '>' not supported between instances of 'NoneType' and 'int'
dispatch_status(20, False, False) -> 'on_time'
dispatch_status(nan, False, False) -> 'missed'
dispatch_status(None, True, False) -> TypeError: '>=' not supported between instances of 'NoneType' and 'int'
```

Step 4: Run the Hypothesis tests against the original script

```
$ python -m pytest test_photo_print_order.py -v --tb=no
============================= test session starts ==============================
platform darwin -- Python 3.15.2, pytest-9.2.0, pluggy-1.7.0 -- /Users/sabiqueislam/Downloads/stairway/.venv/bin/python
hypothesis profile 'default'
rootdir: /Users/sabiqueislam/Downloads/stairway
configfile: pytest.ini
plugins: hypothesis-6.170.15
collecting ... collected 16 items

test_photo_print_order.py::test_within_included_prints PASSED            [  6%]
test_photo_print_order.py::test_order_valid PASSED                       [ 12%]
test_photo_print_order.py::test_extra_print_charge_arbitrary_input FAILED [ 18%]
test_photo_print_order.py::test_extra_print_charge_zero_within_allowance PASSED [ 25%]
test_photo_print_order.py::test_extra_print_charge_never_drops_with_more_copies PASSED [ 31%]
test_photo_print_order.py::test_non_integer_copies_rejected FAILED       [ 37%]
test_photo_print_order.py::test_unknown_finish_rejected FAILED           [ 43%]
test_photo_print_order.py::test_non_bool_membership_rejected FAILED      [ 50%]
test_photo_print_order.py::test_order_valid_arbitrary_input_returns_bool FAILED [ 56%]
test_photo_print_order.py::test_order_valid_matches_limits PASSED        [ 62%]
test_photo_print_order.py::test_order_valid_rejects_bad_file_size FAILED [ 68%]
test_photo_print_order.py::test_order_valid_rejects_bad_copies FAILED    [ 75%]
test_photo_print_order.py::test_dispatch_status_arbitrary_input FAILED   [ 81%]
test_photo_print_order.py::test_dispatch_status_rejects_bad_hours FAILED [ 87%]
test_photo_print_order.py::test_dispatch_status_rejects_non_bool_flags FAILED [ 93%]
test_photo_print_order.py::test_dispatch_status_never_worse_with_more_notice PASSED [100%]

=========================== short test summary info ============================
FAILED test_photo_print_order.py::test_extra_print_charge_arbitrary_input - T...
FAILED test_photo_print_order.py::test_non_integer_copies_rejected - BaseExce...
FAILED test_photo_print_order.py::test_unknown_finish_rejected - Failed: DID ...
FAILED test_photo_print_order.py::test_non_bool_membership_rejected - Failed:...
FAILED test_photo_print_order.py::test_order_valid_arbitrary_input_returns_bool
FAILED test_photo_print_order.py::test_order_valid_rejects_bad_file_size - Ex...
FAILED test_photo_print_order.py::test_order_valid_rejects_bad_copies - Excep...
FAILED test_photo_print_order.py::test_dispatch_status_arbitrary_input - Type...
FAILED test_photo_print_order.py::test_dispatch_status_rejects_bad_hours - Ba...
FAILED test_photo_print_order.py::test_dispatch_status_rejects_non_bool_flags
======================== 10 failed, 6 passed in 1.13s ========================
```

Step 4 (detail): Failing inputs found by Hypothesis

```
$ python -m pytest test_photo_print_order.py -k arbitrary_input --tb=short
============================= test session starts ==============================
platform darwin -- Python 3.15.2, pytest-9.2.0, pluggy-1.7.0
rootdir: /Users/sabiqueislam/Downloads/stairway
configfile: pytest.ini
plugins: hypothesis-6.170.15
collected 16 items / 13 deselected / 3 selected

test_photo_print_order.py FFF                                            [100%]

=================================== FAILURES ===================================
_____________________ test_extra_print_charge_arbitrary_input _____________________
test_photo_print_order.py:38: in test_extra_print_charge_arbitrary_input
    def test_extra_print_charge_arbitrary_input(size, copies, finish, is_member):
                   ^^^
test_photo_print_order.py:40: in test_extra_print_charge_arbitrary_input
    charge = extra_print_charge(size, copies, finish, is_member)
             ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
photo_print_order.py:7: in extra_print_charge
    if copies < 0:
       ^^^^^^^^^^
E   TypeError: '<' not supported between instances of 'NoneType' and 'int'
E   Failing test case: test_extra_print_charge_arbitrary_input(
E       size='4x6',
E       copies=None,
E       finish=None,
E       is_member=None,
E   )
_____________________ test_order_valid_arbitrary_input_returns_bool _____________________
  + Exception Group Traceback (most recent call last):
  |   File "/Users/sabiqueislam/Downloads/stairway/Lab7/test_photo_print_order.py", line 86, in
    test_order_valid_arbitrary_input_returns_bool
  |     def test_order_valid_arbitrary_input_returns_bool(file_mb, copies, is_rush):
  |                    ^^^
  | ExceptionGroup: Hypothesis found 3 distinct failures. (3 sub-exceptions)
  +-+---------------- 1 ----------------
    | Traceback (most recent call last):
    |   File "/Users/sabiqueislam/Downloads/stairway/Lab7/test_photo_print_order.py", line 87, in
    test_order_valid_arbitrary_input_returns_bool
    |     assert isinstance(order_valid(file_mb, copies, is_rush), bool)
    |                       ~~~~~~~~~~~^^^^^^^^^^^^^^^^^^^^^^^^^^^
    |   File "/Users/sabiqueislam/Downloads/stairway/Lab7/photo_print_order.py", line 28, in order_valid
    |     return copies <= 50
    |            ^^^^^^^^^^^
    | TypeError: '<=' not supported between instances of 'NoneType' and 'int'
    | Failing test case: test_order_valid_arbitrary_input_returns_bool(
    |     file_mb=None,
    |     copies=None,
    |     is_rush=True,
    | )
    +---------------- 2 ----------------
    | Traceback (most recent call last):
    |   File "/Users/sabiqueislam/Downloads/stairway/Lab7/test_photo_print_order.py", line 87, in
    test_order_valid_arbitrary_input_returns_bool
    |     assert isinstance(order_valid(file_mb, copies, is_rush), bool)
    |                       ~~~~~~~~~~~^^^^^^^^^^^^^^^^^^^^^^^^^^^
    |   File "/Users/sabiqueislam/Downloads/stairway/Lab7/photo_print_order.py", line 31, in order_valid
    |     return copies <= 500
    |            ^^^^^^^^^^^^
    | TypeError: '<=' not supported between instances of 'NoneType' and 'int'
    | Failing test case: test_order_valid_arbitrary_input_returns_bool(
    |     file_mb=0.0,
    |     copies=None,
    |     is_rush=None,
    | )
    +---------------- 3 ----------------
    | Traceback (most recent call last):
    |   File "/Users/sabiqueislam/Downloads/stairway/Lab7/test_photo_print_order.py", line 87, in
    test_order_valid_arbitrary_input_returns_bool
    |     assert isinstance(order_valid(file_mb, copies, is_rush), bool)
    |                       ~~~~~~~~~~~^^^^^^^^^^^^^^^^^^^^^^^^^^^
    |   File "/Users/sabiqueislam/Downloads/stairway/Lab7/photo_print_order.py", line 29, in order_valid
    |     if file_mb > 25:
    |        ^^^^^^^^^^^
    | TypeError: '>' not supported between instances of 'NoneType' and 'int'
    | Failing test case: test_order_valid_arbitrary_input_returns_bool(
    |     file_mb=None,
    |     copies=None,
    |     is_rush=None,
```

```
    | )
    +----------------------------------
_____________________ test_dispatch_status_arbitrary_input _____________________
test_photo_print_order.py:114: in test_dispatch_status_arbitrary_input
    def test_dispatch_status_arbitrary_input(hours, is_framed, is_international):
                  ^^^
test_photo_print_order.py:116: in test_dispatch_status_arbitrary_input
    status = dispatch_status(hours, is_framed, is_international)
             ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
photo_print_order.py:41: in dispatch_status
    if hours_before_dispatch >= cutoff:
       ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
E   TypeError: '>=' not supported between instances of 'NoneType' and 'int'
E   Failing test case: test_dispatch_status_arbitrary_input(
E       hours=None,
E       is_framed=False,
E       is_international=None,
E   )
========================== short test summary info ===========================
FAILED test_photo_print_order.py::test_extra_print_charge_arbitrary_input - T...
FAILED test_photo_print_order.py::test_order_valid_arbitrary_input_returns_bool
FAILED test_photo_print_order.py::test_dispatch_status_arbitrary_input - Type...
======================== 3 failed, 13 deselected in 0.74s ========================
```

Step 5: Run the fixed script

```
$ python photo_print_order.py
extra_print_charge('4x6', 30, 'matte', False) -> 40
extra_print_charge('4x6', 80, 'canvas', True) -> 770
extra_print_charge('4x6', nan, 'matte', False) -> ValueError: invalid copies
extra_print_charge('4x6', 25.5, 'matte', False) -> ValueError: invalid copies
extra_print_charge('4x6', 25, 'sepia', False) -> ValueError: unknown finish
extra_print_charge(['4x6'], 25, 'matte', False) -> ValueError: unknown print size
extra_print_charge('4x6', None, 'matte', False) -> ValueError: invalid copies
order_valid(10, 100, False) -> True
order_valid(10, -5, False) -> False
order_valid(-1, 10, False) -> False
order_valid(nan, 10, False) -> False
order_valid(None, 10, False) -> False
dispatch_status(20, False, False) -> 'on_time'
dispatch_status(nan, False, False) -> ValueError: invalid hours
dispatch_status(None, True, False) -> ValueError: invalid hours
```

Step 6: Run the Hypothesis tests against the fixed script

```
$ python -m pytest test_photo_print_order.py -v
============================= test session starts ==============================
platform darwin -- Python 3.15.2, pytest-9.2.0, pluggy-1.7.0 -- /Users/sabiqueislam/Downloads/stairway/.venv/bin/python
hypothesis profile 'default'
rootdir: /Users/sabiqueislam/Downloads/stairway
configfile: pytest.ini
plugins: hypothesis-6.170.15
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test_photo_print_order.py::test_dispatch_status_rejects_bad_hours PASSED [ 87%]
test_photo_print_order.py::test_dispatch_status_rejects_non_bool_flags PASSED [ 93%]
test_photo_print_order.py::test_dispatch_status_never_worse_with_more_notice PASSED [100%]

============================== 16 passed in 1.53s ==============================
```